

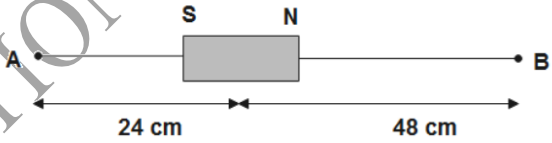
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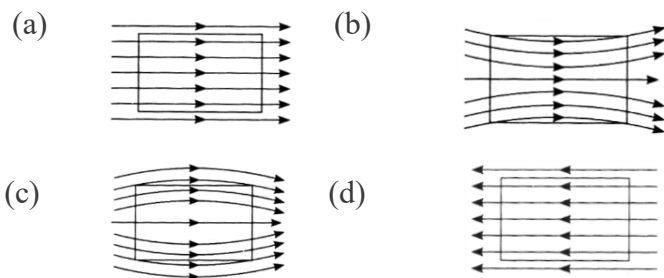
WORKSHEET

PHYSICS

CLASS XII

MAGNETISM AND MATTER

- Two unlike poles of strength 10 Am each are held in air at a distance of 0.10 m from each other. The magnetic force of attraction between them is
(a) 3×10^{-3} N (b) 2.5×10^{-3} N (c) 1.5×10^{-3} N (d) 1×10^{-3} N
- A bar magnet AB of magnetic moment $\vec{m}$ is cut into two equal parts perpendicular to its axis. One part is kept over the other so that end B is exactly over A. The magnetic moment of the combination so formed will be
(a) zero (b) $\vec{m}$ (c) $\vec{m}/4$ (d) $3\vec{m}/4$
- A bar magnet of length 3 cm has points A and B along its axis at distances of 24 cm and 48 cm on the opposite sides. Ratio of magnetic fields at these points will be

(a) 8 (b) $1/2\sqrt{2}$ (c) 3 (d) 4
- The couple acting on a magnet of length 10 cm and pole strength 15 Am, kept in a field of $B = 2 \times 10^{-5}$ T, at an angle of 30° is
(a) 1.5×10^{-5} Nm (b) 1.5×10^{-3} Nm (c) 1.5×10^{-2} Nm (d) 1.5×10^{-6} Nm
- Two wires of same length are shaped into a square and a circle. If they carry same current, ratio of the magnetic moments is
(a) $2: \pi$ (b) $\pi: 2$ (c) $\pi: 4$ (d) $4: \pi$
- Which of the following is weakly repelled by a magnetic field?
(a) Iron (b) Steel (c) Cobalt (d) Copper
- Magnetic permeability is maximum for
(a) ferromagnetic substances (b) diamagnetic substances
(c) paramagnetic substances (d) all of these
- Susceptibility of ferromagnetic substance is
(a) >1 (b) <1 (c) zero (d) 1
- A uniform magnetic field parallel to the plane of paper, existed in space initially directed from left to right. When a bar of soft iron is placed in the field parallel to it, the lines of force passing through it will be represented by figure



10. A bar magnet of magnetic moment 10^4 J/T is free to rotate in a horizontal plane. The work done in rotating the magnet slowly from a direction parallel to a horizontal magnetic field of $4 \times 10^{-5} \text{ T}$ to a direction of 60° from the field will be
 (a) 0.2 J (b) 2 J (c) 4.18 J (d) $2 \times 10^2 \text{ J}$
11. A paramagnetic substance of susceptibility 3×10^{-4} is placed in a magnetizing field of $4 \times 10^4 \text{ A/m}$. Then the intensity of magnetization in the units of A/m is
 (a) 1.33×10^8 (b) 0.75×10^{-8} (c) 12×10^{-8} (d) 14×10^{-8}
12. If the magnetizing field on a ferromagnetic material is increased, its permeability
 (a) is decreased (b) is increased (c) is unaffected (d) may be increased or decreased
13. For a paramagnetic material, the dependence of the magnetic susceptibility χ on the absolute temperature is given as
 (a) $\chi \propto T$ (b) $\chi \propto 1/T^2$ (c) $\chi \propto 1/T$ (d) independent

For questions 14 to 20, two statements are given- one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the options as given below.

- A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.**
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

14. **Assertion:** Magnetic susceptibility is a pure number.
Reason: The value of magnetic susceptibility for vacuum is one.
15. **Assertion:** When radius of a circular loop carrying current is doubled, its magnetic moment becomes four times.
Reason: Magnetic moment depends on the area of the loop.
16. **Assertion:** The poles of magnet cannot be separated by breaking into two pieces.
Reason: The magnetic moment will be reduced to half when a magnet is broken into two equal pieces.
17. **Assertion:** The basic difference between an electric line and magnetic line of force is that former is discontinuous and the latter is continuous or endless.
Reason: No electric lines of forces exist inside a charged body but magnetic lines do exist inside a magnet.
18. **Assertion:** Magnetic moment of an atom is due to both, the orbital motion and spin motion of every electron.
Reason: A charged particle at rest produces a magnetic field.

19. **Assertion:** A paramagnetic sample displays greater magnetization (for the same magnetizing field) when cooled.
Reason: The magnetization does not depend on temperature.
20. **Assertion:** Above the Curie temperature, a ferromagnetic material becomes paramagnetic.
Reason: When a magnetic material is heated to very high temperature, it loses its magnetic properties.

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